MATH 205

The Matrix of a Linear Transformation

LECTURE 16 · THE MATRIX OF A LINEAR TRANSFORMATION

Every gallery picture, every kernel test, every one-to-one check from the last three lectures reduces to one trick. Today we learn it.

SAJAL SOHAIL RANA

WELCOME BACK

The Last Piece

This unit so far

- L13: matrices as functions — a gallery of geometric pictures
- L14: the formal definition, for any vector spaces
- L15: one-to-one, onto, and when T^{-1} exists

Today — closing the loop

- The standard-basis trick: build ANY matrix from a formula
- Composing transformations — and why matrix multiplication works
- Revisiting Lecture 13's gallery with the full toolkit
- Closing worked applications

Course Roadmap

1. Systems of Linear Equations
2. Matrices & Matrix Algebra
3. Determinants
4. Vector Spaces
5. **Linear Transformations**

LECTURE 4 OF 4 — FINAL

After today, Vector Spaces & Linear Transformations is complete — next stop is Inner Products & Gram-Schmidt Process

One formula. Two standard basis vectors. Every matrix you'll ever need.

7.11 THE STANDARD MATRIX

From Formula to Matrix

In Lecture 13, we handed you a matrix and asked what it did. Today we go the other way: given a formula for T, build the matrix.

A Question We've Been Avoiding

Someone hands you a formula:

$$T(x, y) = (x + 2y, \quad 3x - y, \quad x)$$

We PROVED transformations like this are linear back in Lecture 14. But we never showed how to write it as T(x) = Ax. Is there always such a matrix — and how do we find it FAST?

The answer only takes two ingredients you already have: the standard basis vectors e_1, …, e_n, and linearity itself.

Write x in Terms of e_i, Then Apply T

Any x in $\mathbb{R}^n$ breaks apart into standard basis vectors:

$$\underline{x} = x_1\underline{e}_1 + x_2\underline{e}_2 + \cdots + x_n\underline{e}_n$$

Apply T to both sides. Since T is linear, it distributes right through the sum:

$$T(\underline{x}) = x_1T(\underline{e}_1) + x_2T(\underline{e}_2) + \cdots + x_nT(\underline{e}_n)$$

Regroup as a matrix times a vector — the $T(e_i)$'s become COLUMNS:

$$= \begin{bmatrix} T(\underline{e}_1) & T(\underline{e}_2) & \cdots & T(\underline{e}_n) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = A\underline{x}$$

7.11 DEFINITION

The Standard Matrix of T

If T : $\mathbb{R}^n \to \mathbb{R}^m$ is linear, the STANDARD MATRIX of T — sometimes written [T] — is the matrix A with:

$$A = \left[T(\underline{e}_1) \mid T(\underline{e}_2) \mid \cdots \mid T(\underline{e}_n) \right]$$

That's the ENTIRE recipe: compute T on each standard basis vector, and stack the results as columns. No solving systems, no guessing — just n function evaluations.

This is also the reason A is always m×n when T : $\mathbb{R}^n \to \mathbb{R}^m$ — n columns (one per input direction), each with m entries (landing in the output space).

WORKED EXAMPLE

Building the Matrix, Step by Step

$$T(x, y) = (x + 2y, \quad 3x - y, \quad x)$$

Plug in $e_1 = (1,0)$ and $e_2 = (0,1)$ directly:

$$T(\underline{e}_1) = (1, 3, 1), \qquad T(\underline{e}_2) = (2, -1, 0)$$

Assemble as columns:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 1 & 0 \end{bmatrix}$$

WORKED EXAMPLE

Checking It Actually Works

Test on any vector, say x = (2,1). The formula gives T(2,1) = (2+2, 6-1, 2) = (4,5,2). Does Ax match?

$$A\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}4\\5\\2\end{bmatrix} = T(2,1)$$

Matches exactly. This will ALWAYS work — the derivation two slides ago guarantees it for any linear T, not just this example.

Once you have A, everything from Lecture 15 applies immediately: row-reduce for one-to-one/onto, take det(A) for invertibility.

WORKED EXAMPLE

One More, Going Down a Dimension

Let T : $\mathbb{R}^3 \to \mathbb{R}^2$ be given by:

$$T(x, y, z) = (2x - z, \quad x + y + z)$$

Now there are THREE standard basis vectors to plug in — e_1=(1,0,0), e_2=(0,1,0), e_3=(0,0,1):

$T(e_1) = (2,1)$ $T(e_2) = (0,1)$ $T(e_3) = (-1,1)$

(each found by substituting straight into the formula)

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

7.12 COMPOSING TRANSFORMATIONS

Doing Two Things at Once

Rotate, then reflect. Scale, then shear. What matrix does the COMBINED transformation?

The answer is something you did on the course right at the start.

7.12 DEFINITION

Composing Two Transformations

If $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^u$ are linear, their COMPOSITION does T_1 first, then T_2:

$$(T_2 \circ T_1)(\underline{x}) = T_2\big(T_1(\underline{x})\big)$$

Composition of linear transformations is ALWAYS linear (a quick check with the additivity/homogeneity rules confirms it). And its standard matrix is exactly what you'd hope:

$$\text{matrix of } T_2 \circ T_1 \;=\; A_2 A_1$$

Read right to left, matching function notation: apply A_1 first (T_1), then A_2 (T_2) — exactly matrix multiplication order.

WORKED EXAMPLE

Reflect, Then Rotate

Take two transformations straight from Lecture 13's gallery:

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ (reflect } x\text{-axis)}, \qquad A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ (rotate } 90^\circ\text{)}$$

The matrix of "do T_1 first, then T_2" is simply the product A_2A_1:

$$A_2A_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Notice: this combined matrix [[0,1],[1,0]] is exactly reflection about the line y = x from Lecture 13 — reflect-then-rotate-90° turns out to be a single reflection. Composition can simplify in surprising ways.

THE BIG PICTURE

Why Matrix Multiplication Works That Way

Remember learning the matrix multiplication rule — row times column — back in the second week of the course, with no explanation for WHY it's defined that way?

Here's why:

Matrix multiplication is DEFINED to make $(A_2A_1)x$ always equal $A_2(A_1x)$. It's not an arbitrary rule about rows and columns — it's the unique operation that makes “matrix product” agree with “do one transformation, then the other.” Every property of matrix algebra you have memorized — associativity, why order matters ($AB \neq BA$ in general) — is really a fact about composing functions.

Composing transformations in the wrong order gives a different result — exactly why $AB \neq BA$: rotate-then-reflect is generally not the same as reflect-then-rotate.

7.13 BRINGING IT TOGETHER

The Whole Toolkit, at Once

Four lectures of vocabulary — kernel, range, one-to-one, onto, inverse, standard matrix — finally applied together, on transformations you already know by sight.

RECAP

The Gallery, Explained at Last

Every matrix from Lecture 13 was secretly built this exact way — we just hadn't named the trick yet:

Reflection (x-axis)	$T(e_1)=(1,0),\ T(e_2)=(0,-1)$	$[[1,0],[0,-1]]$
Dilation by k	$T(e_1)=(k,0),\ T(e_2)=(0,k)$	$[[k,0],[0,k]]$
Rotation by θ	$T(e_1)=(\cos\theta,\sin\theta),\ T(e_2)=(-\sin\theta,\cos\theta)$	$[[\cos\theta,-\sin\theta],[\sin\theta,\cos\theta]]$
Shear by k	$T(e_1)=(1,0),\ T(e_2)=(k,1)$	$[[1,k],[0,1]]$

Same recipe, every time: evaluate T at e_1 and e_2, write the results as columns.

COMPREHENSIVE EXAMPLE

The Full Pipeline, Start to Finish

$$T(x, y) = (x + y, \quad x - y)$$

1	**Standard matrix**	$T(e_1)=(1,1)$, $T(e_2)=(1,-1)$ → A
2	**One-to-one?**	$\det(A) \neq 0$ ⇒ $\ker(A)=\{0\}$ → YES
3	**Onto?**	A is square + one-to-one ⇒ YES (Lecture 15, square case)
4	**Invertible?**	YES — compute A^{-1} directly

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\det(A) = (1)(-1) - (1)(1) = -2 \neq 0$$

COMPREHENSIVE EXAMPLE

Finishing the Pipeline

Since A is invertible, T^{-1} exists everywhere on $\mathbb{R}^2$ — compute it directly:

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

In four lectures, we went from “what IS a linear transformation” to routinely checking one-to-one, onto, and invertibility for anything given by a formula — all from two standard-basis evaluations and a determinant. That's the entire arc, applied in one slide.

APPLICATION

Building a Graphics Pipeline

A game engine animating a sprite “rotate, then squash vertically” needs exactly ONE combined matrix per frame — not two separate operations:

1	**Rotate 30°**	standard matrix from the rotation formula (Lecture 13)
2	**Scale y by 0.6**	diagonal matrix [[1,0],[0,0.6]]
3	**Combine**	multiply: A_scale · A_rotate — ONE matrix, applied to every vertex

This is exactly why real-time graphics is fast: however many effects are stacked, they collapse into ONE matrix multiply per vertex, computed once per frame rather than once per effect.

UNIT RECAP

Four Lectures, One Story

L13	**Matrices as functions**	domain, codomain, range, and a gallery of geometric pictures
L14	**The formal definition**	$T(u+v)=T(u)+T(v)$, $T(ku)=kT(u)$ — for ANY vector spaces
L15	**One-to-one & Onto**	$\ker(T)=\{0\}$; $R(T)=W$; when T^{-1} exists
L16	**The standard matrix**	$T(e_i)$ as columns; composition = matrix multiplication

RECAP

Today's Key Takeaways

Standard Matrix

$A = [T(e_1) \mid T(e_2) \mid \cdots \mid T(e_n)]$ — the entire recipe.

The Derivation

$x = \Sigma x_i e_i$, apply linearity, regroup — guaranteed to work for any linear T.

Composition

$(T_2 \circ T_1)(x) = T_2(T_1(x))$ is linear, with matrix $A_2 A_1$.

Why Matrix Mult. Works

it's DEFINED so that matrix products match function composition.

The Gallery, Explained

every Lecture 13 matrix came from $T(e_1)$, $T(e_2)$ all along.

Full Pipeline

formula → matrix → one-to-one/onto → invertibility → inverse.

NEXT TIME

Looking Ahead

Before Next Lecture

Build a matrix, then classify it:

For T(x,y,z) = (x+y, y+z, x+2y+z), find the standard matrix, then determine if T is invertible.

MIDTERM II

This whole unit — kernel, range, one-to-one, onto, standard matrix — is fair game for the midterm. Revisit Lectures 13–16 as one connected story, not four separate topics. Make sure you attempt Homework 11.

Lecture 17

Orthogonality & Gram-Schmidt Process

- **Inner Product Spaces: Definition & Axioms**
- **Orthogonality, Projections & Gram-Schmidt Process**
- **Gram-Schmidt Example, Orthogonal Sets & Cauchy-Schwarz**

Questions?

The Matrix of a Linear Transformation · Lecture 16 complete

LINEAR TRANSFORMATIONS MODULE — COMPLETE

NEXT: INNER PRODUCT SPACES & ORTHOGONALITY